

The Complete Positive-Density Riemann Atlas for One-Dimensional Isothermal Euler Flow

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Abstract

For one-dimensional isothermal Euler flow at sound speed $a > 0$, we reduce every positive-density Riemann problem to one strictly monotone scalar equation. Its unique root yields the intermediate density and all four shock/rarefaction patterns with exact fan profiles and Rankine–Hugoniot speeds. We prove the Lax inequalities, strict mechanical-entropy selection and the absence of vacuum for every finite velocity jump.

1 One scalar equation for the full solver

Let $U = (\rho, m)$, $m = \rho u$, and

$$U_t + F(U)_x = 0, \quad F(U) = \left(m, \frac{m^2}{\rho} + a^2 \rho \right), \quad a > 0, \quad \rho > 0. \quad (1)$$

The characteristic speeds are $\lambda_1 = u - a$ and $\lambda_2 = u + a$. For $r, r_0 > 0$ define the continuous C^1 , strictly increasing function

$$f(r; r_0) = \begin{cases} a \log(r/r_0), & 0 < r \leq r_0, \\ a(r - r_0)/\sqrt{rr_0}, & r \geq r_0. \end{cases} \quad (2)$$

Theorem 1 (Complete positive-density Riemann solver). *For arbitrary left and right states $(\rho_L, u_L), (\rho_R, u_R)$ with positive density and finite velocity, there is a unique $\rho_* > 0$ satisfying*

$$f(\rho_*; \rho_L) + f(\rho_*; \rho_R) + u_R - u_L = 0. \quad (3)$$

The unique intermediate velocity is

$$u_* = u_L - f(\rho_*; \rho_L) = u_R + f(\rho_*; \rho_R). \quad (4)$$

The 1-wave from L to $$ is a rarefaction, zero wave, or Lax shock as $\rho_* < \rho_L, =$, or $>$; the 2-wave from $*$ to R has the analogous classification relative to ρ_R . These waves and the constant intermediate sector form the unique self-similar Lax entropy solution. In particular, no finite datum in this chamber produces vacuum.*

Proof. For fixed r_0 , the two derivatives of $f(\cdot; r_0)$ are a/r and $a(r + r_0)/(2r\sqrt{rr_0})$, respectively. They are positive and meet at a/r_0 ; moreover f ranges from $-\infty$ to $+\infty$. Hence the left side of (3) has exactly one positive zero; (4) is then forced. The logarithmic branch is obtained by integrating $du = \mp a \, d\rho/\rho$ on the two integral curves. Eliminating the shock speed from the Rankine–Hugoniot equations gives

$$(u - u_0)^2 = a^2 \frac{(\rho - \rho_0)^2}{\rho \rho_0}. \quad (5)$$

The compressive sign in family 1 is negative and that in family 2 is positive, which gives the second branch of (2). The explicit formulas and entropy check below complete the construction. Positivity of the unique root proves the no-vacuum assertion. $\square$

2 The four patterns in one table

The two comparisons with the unique root give the full atlas.

comparison	family	wave
$\rho_* < \rho_L$	1	rarefaction
$\rho_* > \rho_L$	1	shock
$\rho_* < \rho_R$	2	rarefaction
$\rho_* > \rho_R$	2	shock

Equality deletes the wave continuously. Thus the two independent signs produce rarefaction–rarefaction, rarefaction–shock, shock–rarefaction, and shock–shock solutions without separate existence arguments.

For a 1-rarefaction, on $u_L - a \leq \xi \leq u_* - a$,

$$u(\xi) = \xi + a, \quad \rho(\xi) = \rho_L \exp\left(\frac{u_L - u(\xi)}{a}\right). \quad (6)$$

For a 2-rarefaction, on $u_* + a \leq \xi \leq u_R + a$,

$$u(\xi) = \xi - a, \quad \rho(\xi) = \rho_R \exp\left(\frac{u(\xi) - u_R}{a}\right). \quad (7)$$

If $r_L = \rho_*/\rho_L > 1$, the 1-shock speed is

$$s_1 = u_L - a\sqrt{r_L} = u_* - \frac{a}{\sqrt{r_L}}. \quad (8)$$

If $r_R = \rho_*/\rho_R > 1$, the 2-shock speed is

$$s_2 = u_R + a\sqrt{r_R} = u_* + \frac{a}{\sqrt{r_R}}. \quad (9)$$

3 Lax ordering and mechanical entropy

Equations (8)–(9) give, by direct subtraction,

$$u_* - a < s_1 < u_L - a, \quad u_R + a < s_2 < u_* + a, \quad (10)$$

whenever the corresponding shock exists. The fan speeds in (6)–(7) increase from left to right. Every 1-wave lies below u_* and every 2-wave lies above u_* , so their ordering is strict.

In conservative variables a strictly convex entropy pair is

$$\eta(\rho, m) = \frac{m^2}{2\rho} + a^2 \rho \log \rho, \quad q(\rho, m) = u(\eta + a^2 \rho). \quad (11)$$

Indeed, $\eta_{mm} = 1/\rho > 0$ and $\det D^2 \eta = a^2/\rho^2 > 0$. Smooth fans satisfy equality in the entropy law. For either shock, let $r > 1$ be the compressed-to-outer density ratio and let ρ_0 be that outer, lower density. Direct substitution gives

$$[q] - s[\eta] = a^3 \rho_0 \sqrt{r} \left[\log r - \frac{1}{2}(r - r^{-1}) \right] < 0. \quad (12)$$

Indeed, $h(r) = \frac{1}{2}(r - r^{-1}) - \log r$ obeys $h(1) = 0$ and $h'(r) = (r - 1)^2/(2r^2) > 0$. Thus the compressive signs selected in (10) are strictly entropy admissible; the opposite algebraic signs are expansive and violate the Lax condition.

Every self-similar Lax solver with the prescribed family order must follow these exhaustive 1- and 2-wave curves. Their unique intersection from Theorem 1, together with the strict wave ordering above, proves uniqueness of the assembled two-wave entropy solution.

The same monotonicity that constructs ρ_* gives a useful global fact. Even when $u_R - u_L$ is arbitrarily large, the left side of (3) reaches it only at a strictly positive root. Unlike the $\gamma > 1$ polytropic case, the logarithm diverges at zero and prevents a finite-data vacuum gap.

Literature ownership and AI-use statement

The characteristic, shock and Riemann-solver framework belongs to the classical theory of hyperbolic conservation laws and gas dynamics [1, 2, 3]. We claim no literary priority; this source-local paper derives the frozen isothermal case completely. AI tools assisted algebraic checking, hostile boundary review and manuscript preparation. Every theorem and computation is exposed for independent verification.

References

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